

2.10.27

AI25BTECH11038 – Tejas Uppala

Question:

The scalar $\mathbf{A} \cdot ((\mathbf{B} + \mathbf{C}) \times (\mathbf{A} + \mathbf{B} + \mathbf{C}))$ equals

- | | |
|---------------------|--------------------------------------|
| 1) 0 | 3) $[\mathbf{ABC}] + [\mathbf{BCA}]$ |
| 2) $[\mathbf{ABC}]$ | 4) None of these |

Solution:

Using the **distributive property** of cross product over addition $((\mathbf{B} + \mathbf{C}) \times (\mathbf{A} + \mathbf{B} + \mathbf{C}))$ can be expanded to,

$$((\mathbf{B} + \mathbf{C}) \times (\mathbf{A} + \mathbf{B} + \mathbf{C})) = ((\mathbf{B} + \mathbf{C}) \times \mathbf{A}) + ((\mathbf{B} + \mathbf{C}) \times (\mathbf{B} + \mathbf{C})) \quad (1)$$

The cross product of two parallel vectors is a zero vector,

$$((\mathbf{B} + \mathbf{C}) \times (\mathbf{A} + \mathbf{B} + \mathbf{C})) = ((\mathbf{B} + \mathbf{C}) \times \mathbf{A}) \quad (2)$$

$$((\mathbf{B} + \mathbf{C}) \times (\mathbf{A} + \mathbf{B} + \mathbf{C})) = (\mathbf{B} \times \mathbf{A}) + (\mathbf{C} \times \mathbf{A}) \quad (3)$$

Now,

$$\mathbf{A} \cdot ((\mathbf{B} + \mathbf{C}) \times (\mathbf{A} + \mathbf{B} + \mathbf{C})) = \mathbf{A} \cdot ((\mathbf{B} \times \mathbf{A}) + (\mathbf{C} \times \mathbf{A})) \quad (4)$$

$$\mathbf{A} \cdot ((\mathbf{B} + \mathbf{C}) \times (\mathbf{A} + \mathbf{B} + \mathbf{C})) = (\mathbf{A} \cdot (\mathbf{B} \times \mathbf{A})) + (\mathbf{A} \cdot (\mathbf{C} \times \mathbf{A})) \quad (5)$$

Since the two vectors $\mathbf{A}$ and $\mathbf{B} \times \mathbf{A}$ are perpendicular and so are $\mathbf{A}$ and $\mathbf{C} \times \mathbf{A}$, their dot product is zero

$$\mathbf{A} \cdot ((\mathbf{B} + \mathbf{C}) \times (\mathbf{A} + \mathbf{B} + \mathbf{C})) = 0 \quad (6)$$